

Group 16: Melanoma Detection from Dermoscopic Images Using Deep Learning and Cross-Dataset Evaluation

Kyle Packer

Ann Nguyen

Artien Voskanian

Whitney deRoche

Abstract

Melanoma is the deadliest form of skin cancer, and early detection is critical for improving patient outcomes. In this project, we investigate automated melanoma detection from dermoscopic images using machine learning and deep learning methods trained on the ISIC 2020 skin lesion dataset and evaluated both in-domain and on the external ISIC 2019 dataset [8, 7]. Our input consists primarily of dermoscopic images, with additional experiments exploring patient metadata fusion, and the output is a binary prediction of melanoma versus non-melanoma. We implemented a Random Forest baseline using handcrafted lesion features, along with transfer-learning approaches based on ResNet-50 and EfficientNet-B4, using normalization, padding, augmentation, and class-imbalance handling during training. While the deep learning models achieved stronger in-domain validation performance, external evaluation on ISIC 2019 revealed reduced generalization, underscoring the importance of domain shift in medical image classification. Overall, our results suggest that pretrained convolutional models are effective for melanoma screening, but robustness across datasets remains a key challenge for reliable clinical deployment.

1 Introduction

Melanoma poses a major diagnostic challenge in dermatology because early-stage detection is closely tied to improved patient outcomes. In clinical practice, dermatologists often rely on dermoscopy, a noninvasive imaging technique that reveals subsurface skin structures and visual patterns associated with malignant lesions. However, diagnosing melanoma from dermoscopic images remains challenging because lesions can vary substantially in color, texture, border irregularity, and overall appearance. This makes melanoma detection an important and well-motivated problem for machine learning, particularly for computer vision systems that can assist clinicians by identifying subtle image patterns and improving consistency in screening workflows.

In this project, we study automated melanoma detection using dermoscopic skin lesion images from the ISIC 2019 and ISIC 2020 challenge datasets. Our primary predictive

task is binary classification: Given a dermoscopic image of a skin lesion, the model outputs a probability estimate indicating the likelihood that the lesion is melanoma versus non-melanoma. In some experiments, we also explore the use of patient metadata alongside the image input. We compare several approaches to this task, beginning with a Random Forest baseline built on handcrafted lesion descriptors inspired by clinically meaningful properties such as asymmetry and color variation, and then moving to deep learning models based on transfer learning with ResNet-50 and EfficientNet-B4. These neural models take normalized dermoscopic images as input and produce a probability score for melanoma, allowing us to evaluate performance under different classification thresholds.

A major challenge in this problem is class imbalance and generalization. In the ISIC 2020 training data, melanoma cases make up only a small fraction of the total dataset, making naive classification approaches prone to favoring the majority class. To address this, our models incorporate imbalance-aware training strategies such as weighted sampling and loss functions designed for rare positive cases. At the same time, strong in-domain validation performance does not necessarily imply reliable real-world performance. Because ISIC 2020 does not provide public test labels, we use ISIC 2019 as an external evaluation dataset to measure cross-dataset generalization. This allows us to examine an issue that is especially important in medical machine learning: domain shift. Differences in acquisition conditions, label distributions, and dataset composition can significantly reduce performance when a model is deployed outside the environment in which it was trained.

Our project is motivated not only by the promise of high-performing image classifiers, but also by the need to understand how different modeling choices affect robustness. We therefore evaluate traditional feature-based methods, modern convolutional neural networks, and later extensions such as ensembling and metadata fusion. Through this comparison, we evaluate whether more expressive models improve melanoma detection in a meaningful and generalizable way. Overall, our results highlight both the potential and the limitations of deep learning for medical image analysis: while pretrained vision models can achieve strong results on dermoscopic classification tasks, external validation remains essential before such systems can be considered reliable for clinical support.

2 Related Work

Skin lesion classification has traditionally been performed by dermatologists through visual inspection and dermoscopy, but this process can be subjective and difficult, especially when melanoma resembles benign lesions. Earlier computational approaches relied on handcrafted features such as asymmetry, border irregularity, color variation, and lesion diameter, followed by classical machine learning models. These methods were appealing because they were relatively interpretable and reflected the visual criteria used by dermatologists, but they depended heavily on manual feature design and often lacked robustness across datasets. The development of public benchmark challenges also played an important role in advancing the field; for example, the ISIC 2017 challenge formalized lesion analysis tasks and helped standardize comparison across competing approaches [2]. Our Random Forest baseline is most similar to this earlier line of work, although our project goes further by comparing handcrafted-feature methods against modern deep learning models under external evaluation.

The field changed substantially with deep convolutional neural networks. Esteva and colleagues showed that transfer learning on large image datasets could produce skin cancer classifiers with dermatologist-level performance, helping establish end-to-end CNNs as a dominant approach for this task [3]. That work was especially influential because it demonstrated that skin lesion diagnosis could be treated as a large-scale image classification problem rather than requiring extensive manual feature engineering. In practice, this line of work has become the foundation for many later systems, including our own ResNet-50 and EfficientNet-based models.

A major branch of later work focuses on stronger backbone architectures. ResNet, introduced by He and colleagues, used residual connections to enable more effective training of very deep neural networks [5]. This was a foundational contribution because it addressed the optimization difficulty that often comes with increasing depth, and ResNet variants remain common baselines in medical imaging. The EfficientNet model introduced by Tan and Le uses compound scaling, where depth, width, and input resolution are increased together in a coordinated way [27]. This scaling strategy is especially compelling because it improves both accuracy and efficiency without requiring a more complex training pipeline. That helps explain why EfficientNet-style models have become attractive for skin lesion tasks, where researchers often want strong performance without excessive computational cost.

Recent application papers build on these architecture ideas in different ways. For example, SBXception modifies Xception into a shallower and broader network and reports strong performance on HAM10000 while also reducing parameter count and training time [12, 29]. This is an

interesting design choice because it challenges the assumption that deeper is always better. However, the study also notes the need for broader validation beyond a single benchmark dataset, which is important because strong in-dataset accuracy does not always translate into reliable external performance [6].

Another important category is ensemble learning. Selvaraj et al. combine several modern CNN backbones, including ResNeXt101, SeResNeXt101, ResNet152V2, DenseNet201, GoogLeNet, and Xception, and show that weighted ensembling can outperform individual models, reaching a macro-average ROC-AUC of 0.97 [26]. This is a strong and sensible approach because different backbones capture different visual patterns, and combining them can reduce model variance and improve robustness. A key limitation is that ensemble methods are computationally heavier and can be harder to interpret or deploy in low-resource clinical settings.

A particularly relevant recent direction is multimodal learning, where image features are combined with meta-data such as age, sex, and anatomical site. Saeed et al. propose a multimodal deep learning ensemble framework that fuses metadata with image features from pretrained CNNs and then applies adaptive weighted ensembling [20]. This approach is especially close to our project because we also explored metadata-enhanced classification. The motivation is clear: melanoma risk is not purely visual, so adding patient context can improve discrimination. Their results suggest that metadata fusion improves accuracy, precision, recall, and AUC on ISIC datasets, and the model also retains competitive performance on an external Derm7pt dataset [20]. At the same time, the paper still reports a noticeable drop from internal validation to external testing, which highlights a central issue in this literature: even sophisticated multimodal systems remain vulnerable to dataset shift.

Overall, the state of the art in skin lesion classification is no longer classical feature engineering or unaided manual inspection. The strongest current methods are deep transfer-learning systems built on architectures such as ResNet, EfficientNet, and Xception, increasingly strengthened by ensembling and metadata fusion [3, 5, 27, 12, 26, 20]. Among these, EfficientNet-style scaling stands out for improving model efficiency in a principled way, while multimodal ensemble learning stands out for incorporating both lesion appearance and patient context. However, the literature also suggests that benchmark performance alone is not enough. Many studies report excellent internal accuracy or AUC, but external validation remains a harder and more informative test. That is where our project differs most clearly from much of the prior work: rather than evaluating only in-domain success, we compare classical and deep models while explicitly examining melanoma-specific performance and generalization from ISIC 2020 to ISIC 2019.

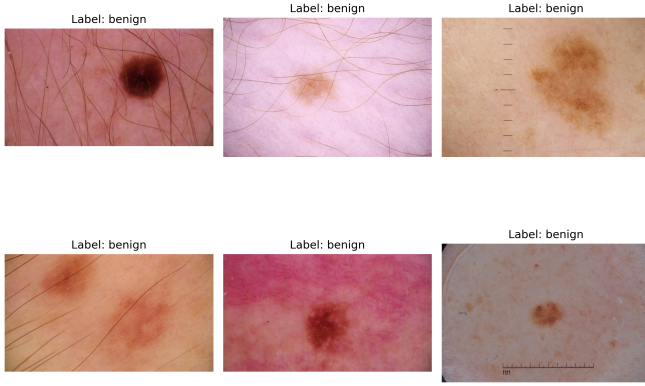


Figure 1: Representative dermoscopic images from the ISIC dataset used in this study.

3 Dataset and Features

We trained and evaluated our models using dermoscopic skin lesion images from the ISIC 2020 and ISIC 2019 challenge datasets, obtained from the International Skin Imaging Collaboration (ISIC) archive [8, 7, 19]. For model development, we used the ISIC 2020 training dataset, which contains 33,126 JPEG images with associated lesion diagnoses and metadata fields such as patient age, sex, and general anatomic site. Of these, 6,626 images were held out to form a validation set. Because ISIC 2020 does not provide public test labels, we used the ISIC 2019 test dataset as an external evaluation benchmark; this dataset contains 8,238 JPEG dermoscopic images with gold-standard labels and accompanying metadata. These datasets are widely used in skin lesion analysis and provide a realistic setting for studying melanoma detection under class imbalance and dataset shift.

Our primary predictive task was binary classification of melanoma versus non-melanoma from dermoscopic images. However, the ISIC 2019 and ISIC 2020 datasets could also be used for several other predictive tasks, including multi-class lesion classification and metadata-aware melanoma risk prediction using variables such as age, sex, and general anatomic site alongside image features [8, 7]. We chose binary melanoma detection because it most directly reflects the clinical importance of early melanoma screening and provides a focused setting for evaluating model sensitivity to rare but high-risk cases. This formulation also allowed us to concentrate on the most important technical issues in the dataset, especially class imbalance and domain shift, rather than distributing effort across a larger multiclass label space. The ISIC 2020 training set is highly imbalanced, with only about 2% melanoma cases, making standard accuracy-based evaluation insufficient and motivating the use of ROC-AUC and especially PR-AUC. Representative examples of dermoscopic images from the dataset are shown in Figure 1.

Before training, we applied several preprocessing steps to

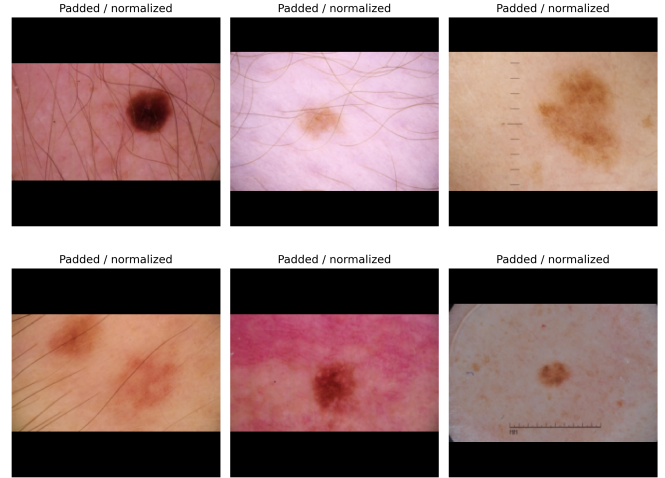


Figure 2: Example dermoscopic images after padding-based resizing and normalization, illustrating the standardized model inputs.

standardize the images while preserving lesion structure. Since the original dermoscopic images vary substantially in size and aspect ratio, we used padding-based resizing rather than direct warping so that images could be converted to fixed square inputs without distorting lesion shape. We then normalized the RGB channels using standard ImageNet mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$ values so that the inputs matched the expectations of pretrained convolutional backbones. For data augmentation, we applied random horizontal and vertical flips, random rotations, and mild color jitter to improve robustness to orientation and illumination differences. In our experiments, ResNet-50 used 224×224 inputs, while EfficientNet-B4 used 380×380 inputs, following the respective pretrained model conventions.

Examples of images after padding-based resizing and normalization are shown in Figure 2, illustrating the standardized inputs used for model training.

We explored two main types of features in this project. First, for our classical machine learning baseline, we extracted handcrafted lesion descriptors inspired by dermatologic criteria such as asymmetry and color variation. These features included summary statistics of color distributions in RGB and LAB color spaces, including mean, standard deviation, skewness, kurtosis, and entropy, as well as multi-axis asymmetry metrics designed to capture irregular lesion structure. These handcrafted features were used in a Random Forest classifier. Second, for our deep learning models, we used the raw dermoscopic images directly as input, allowing pretrained convolutional neural networks to learn hierarchical texture, color, and shape features automatically from the data. In later modeling stages, we also explored metadata fusion by incorporating structured variables such as age, sex, and anatomic site

alongside the image features.

4 Method

4.1 Baseline - Random Forest

As a classical machine learning baseline, a Random Forest classifier was trained on 49 hand-engineered features extracted from raw dermoscopic images, grounded in established clinical criteria for melanoma detection [1, 25, 10]. Features fell into two categories motivated by the ABCDE rule: color analysis and asymmetry quantification. For color, 39 features were extracted across RGB and LAB color spaces — per-pixel normalized RGB channels each yielded nine statistics (mean, standard deviation, skewness, kurtosis, range, 25th/50th/75th percentiles, and entropy), while the LAB a and b channels contributed mean, standard deviation, and range, exploiting LAB’s perceptual uniformity for capturing color variegation [13]. For asymmetry, 10 features were extracted by comparing image halves and opposite quadrant pairs across horizontal, vertical, and diagonal axes on grayscale images, computing mean, standard deviation, and maximum absolute differences per axis alongside a composite asymmetry score. The ISIC 2020 data used for the Random Forest baseline were split into 80/20 training and validation sets with stratification to preserve class distribution across splits. The Random Forest was configured with 300 estimators, a maximum depth of 10, and balanced class weights to mitigate bias toward the majority benign class, a persistent challenge given the severe class imbalance in dermoscopic datasets. Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

4.2 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a class of deep learning models designed for structured grid data such as images. Unlike fully connected networks, CNNs utilize spatial locality by applying learned filters across the input via the convolution operation. For a 2D input image $X \in \mathbb{R}^{H \times W \times C}$ and a filter $K \in \mathbb{R}^{k_h \times k_w \times C}$, the convolution output at location (i, j) is:

$$Y_{i,j} = \sum_{u=0}^{k_h-1} \sum_{v=0}^{k_w-1} \sum_{c=0}^{C-1} K_{u,v,c} \cdot X_{i+u,j+v,c} + b$$

where k_h, k_w are the kernel dimensions, C is the number of input channels, and b is a bias term. This operation is applied across multiple channels and repeated across many layers, allowing the network to learn hierarchical features — from low-level edges and textures in early layers to high-level semantic representations in deeper layers. Each convolutional layer is typically followed by

a nonlinear activation function such as ReLU, defined as $\text{ReLU}(z) = \max(0, z)$, which introduces nonlinearity while mitigating the vanishing gradient problem. The final layer produces a class probability via the sigmoid function for binary classification:

$$\hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

where $\hat{p} \in (0, 1)$ represents the predicted probability of the positive class [9].

4.3 ResNet50

ResNet50 is a deep convolutional neural network consisting of 50 layers organized into residual blocks. A core challenge in training very deep networks is the *vanishing gradient problem*, where gradients shrink exponentially during backpropagation, making early layers difficult to train. ResNet addresses this issue by introducing *skip connections* that allow gradients to flow directly through the network.

For an input $\mathbf{x}$ to a residual block, the output is defined as

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (1)$$

where $\mathcal{F}(\mathbf{x}, \{W_i\})$ represents the residual mapping learned by the stacked layers (typically two or three convolutional layers with batch normalization and ReLU), and $\mathbf{x}$ is the identity shortcut connection. Rather than learning the full desired mapping $H(\mathbf{x})$, the network instead learns the residual function $\mathcal{F}(\mathbf{x}) = H(\mathbf{x}) - \mathbf{x}$, which is empirically easier to optimize [5].

In this work, ResNet50 was initialized with ImageNet pretrained weights and fine-tuned on the ISIC 2020 training set for binary melanoma classification. During Stage 1, the backbone weights were frozen and only the classification head was trained for 5 epochs using the Adam optimizer with a learning rate of 1×10^{-3} , a `WeightedRandomSampler` with a $5\times$ oversampling ratio for positive class, and Focal Loss with $\alpha = 0.71$ and $\gamma = 2$.

In Stage 2, the backbone was unfrozen for end-to-end fine-tuning over an additional 5 epochs with a reduced learning rate of 1×10^{-5} , allowing the model to adapt pretrained feature representations to the dermoscopy domain.

4.4 EfficientNet-B4

EfficientNet-B4 is a convolutional neural network architecture that scales network depth, width, and input resolution jointly using a compound scaling method. Rather than arbitrarily increasing a single dimension, EfficientNet-B4

applies a compound coefficient ϕ to uniformly scale all three dimensions according to

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi \quad (2)$$

subject to the constraint $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$, where d , w , and r denote the depth, width, and resolution scaling factors, respectively. This principled scaling approach yields strong performance while maintaining parameter efficiency compared to conventional CNN architectures [27].

In this work, EfficientNet-B4 was initialized with ImageNet pretrained weights and trained end-to-end for 10 epochs using the Adam optimizer with a learning rate of 1×10^{-4} . Unlike ResNet50, EfficientNet-B4 was trained in a single stage without freezing the backbone.

To address class imbalance, a positive class weight w^+ was computed using balanced class weighting and incorporated into the Binary Cross-Entropy with Logits loss:

$$\mathcal{L} = -[w^+ \cdot y \log \sigma(z) + (1 - y) \log (1 - \sigma(z))], \quad (3)$$

where y is the ground truth label and z is the raw model logit. Mixed precision training was employed via gradient scaling to improve numerical stability and GPU throughput.

4.5 Ensemble Construction

Two convolutional neural network architectures, ResNet50 and EfficientNet, were independently trained on the ISIC 2020 dataset and served as base models for the ensemble. To determine the optimal method of combining their outputs, three candidate fusion strategies were evaluated on the held-out validation set: *logit averaging*, in which the raw pre-sigmoid outputs of both models were averaged prior to activation [4]; *rank averaging*, in which each model’s predictions were converted to percentile ranks before being averaged across models; and *probability averaging*, in which the sigmoid-transformed output probabilities of each model were directly averaged. The fusion strategy yielding the highest validation AUC was selected for downstream evaluation.

To reduce prediction variance and improve generalization, three augmentation strategies were evaluated in conjunction with the selected fusion method: test-time augmentation (TTA) alone, geometric lesion augmentation alone, and a combined TTA + geometric lesion augmentation condition.

4.5.1 Geometric Lesion Augmentation:

Each lesion was first *center-cropped* to a specified fraction of the original image size (default 90%) and then resized back to the original spatial dimensions using bilinear interpolation. This ensures that the model focuses on the lesion while maintaining consistent input size. Following center cropping, spatial transformations calibrated to lesion morphology were applied, including random affine deformations and perspective distortions, further diversifying the input images.

4.5.2 Test-Time Augmentation (TTA):

TTA was implemented by generating multiple transformed views of each test image, depending on the selected augmentation mode:

- *none*: no augmentation.
- *light*: horizontal flip only.
- *basic*: horizontal flip, vertical flip, and 180° rotation.
- *eight*: basic augmentations plus 90° and 270° rotations and flipped versions of these rotations.

Predictions were obtained by averaging the sigmoid probabilities across all TTA views.

The best-performing combination of fusion strategy and augmentation condition, as determined by validation AUC, was subsequently applied to the held-out ISIC 2019 test set to assess cross-domain generalization. No further tuning or selection was performed after test set evaluation to avoid data leakage.

4.6 Libraries

All experiments were implemented in Python (version 3.11). Numerical computations and data handling were performed using NumPy [14] and Pandas [15], while statistical analyses relied on SciPy [23]. Machine learning algorithms, including PCA and Random Forest, were implemented with scikit-learn [22]. Deep learning models were developed using PyTorch [17] and Torchvision [18], with access to additional architectures through the timm library [30]. Image processing and manipulation were performed using Pillow [16] and scikit-image [21]. Data visualization was conducted using Matplotlib [11] and Seaborn [24], and progress tracking during experiments was enabled by tqdm [28].

5 Experiments/Results/Discussion

This section describes the experimental setup, evaluation metrics, and the results obtained from our baseline, deep learning, and ensemble models. All model development and hyperparameter tuning were performed using the ISIC 2020 training dataset with a held-out validation set, while the ISIC 2019 dataset was reserved exclusively for final external evaluation in order to assess cross-dataset generalization.

5.1 Experimental Setup

For model development, the ISIC 2020 training data were divided into training and validation sets using an 80/20 stratified split. Because melanoma cases represent only a small fraction of the dataset, evaluation metrics robust to class imbalance were emphasized.

ResNet50 and EfficientNet-B4 were trained using the procedures described in Section 4, with model selection performed on a held-out validation split from ISIC 2020.

Input preprocessing followed the requirements of the pre-trained architectures. ResNet50 used images resized with padding to 224×224 , while EfficientNet-B4 used 380×380 inputs. In both cases, images were normalized using ImageNet mean and standard deviation values. Data augmentation included random flips, rotations, and color jitter to improve robustness to orientation and illumination differences.

Model performance was primarily evaluated using ROC-AUC and PR-AUC, which are well suited to highly imbalanced classification tasks. The ROC-AUC metric measures the model’s ability to rank positive examples above negative ones across all possible classification thresholds. It can be expressed as

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx$$

where TPR is the true positive rate and FPR is the false positive rate. Because melanoma cases are rare, we also emphasize the Precision-Recall AUC, which focuses on performance for the positive class.

5.2 Baseline Model

We first evaluated a Random Forest classifier trained on handcrafted lesion descriptors derived from dermoscopic images. These features included color distribution statistics (mean, variance, skewness, kurtosis, and entropy) across multiple color spaces, as well as asymmetry metrics designed to capture irregular lesion structure.

The Random Forest achieved an accuracy of approximately 0.82, with melanoma precision and recall both around 0.43, and an ROC-AUC of approximately 0.84 on the validation set. While these results indicate that engineered features contain useful information for ranking lesions, the relatively modest classification metrics suggest that handcrafted features alone are limited in their ability to cleanly separate melanoma from benign lesions at a fixed threshold. This baseline therefore serves primarily as a point of comparison for the more expressive deep learning models.

5.3 CNN Model Performance

Table 1 summarizes validation performance for the deep learning models. Both convolutional neural networks outperform the handcrafted baseline in terms of ranking performance.

Model	ROC-AUC	PR-AUC
Random Forest	0.84	N/A
ResNet50	0.887	0.188
EfficientNet-B4	0.909	0.211

Table 1: Validation performance of the baseline and image-only deep learning models.

EfficientNet-B4 achieved the strongest individual model performance, with a validation ROC-AUC of 0.909 and PR-AUC of 0.211, slightly outperforming ResNet50. This result is consistent with prior work suggesting that EfficientNet’s compound scaling strategy allows the model to capture richer visual patterns while maintaining computational efficiency.

Although both networks performed well individually, their predictions were not identical. The correlation between their validation predictions was approximately 0.54, indicating that the models learned partially complementary representations. This observation motivated the use of ensemble learning.

5.4 Ensemble Experiments

We evaluated three ensemble strategies for combining the predictions of ResNet50 and EfficientNet-B4:

- probability averaging
- rank averaging
- logit averaging

Each strategy was tested across several augmentation settings, including no augmentation, test-time augmentation (TTA), lesion-focused cropping, and a combined cropping

Experiment	Best Method	ROC-AUC	PR-AUC
Baseline	Logit	0.9197	0.2278
TTA only	Logit	0.9220	0.2404
Crop only	Rank	0.9089	0.1883
Crop + TTA	Rank	0.9140	0.1936

Table 2: Best ensemble result from each validation experiment.

+ TTA configuration. Table 2 summarizes the best ensemble result from each experiment.

The best overall configuration used logit averaging with test-time augmentation, achieving a validation ROC-AUC of 0.922 and PR-AUC of 0.240. In this configuration, the optimal ensemble weighting assigned approximately 0.73 weight to ResNet50 and 0.27 to EfficientNet-B4. This improvement over the individual models suggests that combining architectures with partially independent errors can reduce prediction variance and improve overall ranking performance.

5.5 External Evaluation and Domain Shift

After selecting the best validation configuration, the ensemble was evaluated on the ISIC 2019 test dataset, which was not used during training or hyperparameter tuning. This evaluation provides insight into cross-dataset generalization. Table 3 summarizes the external test results.

Model	ROC-AUC	PR-AUC
ResNet50	0.711	0.379
EfficientNet-B4	0.710	0.368
Ensemble (Logit + TTA)	0.723	0.390

Table 3: External evaluation on the ISIC 2019 test dataset.

The ensemble again achieved the strongest performance, demonstrating that the fusion strategy improves results even on unseen data. However, a clear performance drop is observed when comparing validation and external evaluation results. The validation ROC-AUC of approximately 0.92 decreases to 0.72 on the ISIC 2019 dataset.

This decline provides empirical evidence of domain shift between the ISIC 2020 and ISIC 2019 datasets. The observed performance drop suggests that differences between the ISIC 2020 and ISIC 2019 datasets, such as variations in image acquisition, lesion characteristics, or dataset composition, affect the model’s ability to generalize.

5.6 Metadata Experiments

We also explored whether incorporating patient metadata could improve performance. Structured features including age, sex, and anatomical site were concatenated with image-based representations for both ResNet50 and EfficientNet-B4 models. However, external evaluation revealed that metadata integration actually degraded performance.

Model	ROC-AUC	PR-AUC
ResNet50 + Metadata	0.639	0.239
EfficientNet-B4 + Metadata	0.661	0.288

Table 4: External evaluation of image-plus-metadata models on the ISIC 2019 test dataset.

This result likely reflects differences in metadata distributions between the ISIC 2020 and ISIC 2019 datasets, which may cause the model to learn dataset-specific correlations that do not generalize. As a result, although multimodal approaches can be powerful, they may also increase susceptibility to dataset shift.

Overall, these experiments highlight two key findings. First, deep convolutional neural networks substantially outperform handcrafted features for melanoma detection, and combining architectures through ensembling further improves predictive performance. Second, the large gap between validation and external test performance demonstrates that even high-performing models remain sensitive to dataset shift. This emphasizes the need for rigorous external validation when developing medical image classification systems intended for real-world deployment.

6 Conclusion

In this work, we investigated automated melanoma detection from dermoscopic images using both classical machine learning and deep learning approaches. A Random Forest baseline using handcrafted lesion descriptors demonstrated that engineered features contain useful signal but are limited in their ability to reliably distinguish melanoma from benign lesions. Deep convolutional neural networks substantially improved ranking performance, with EfficientNet-B4 slightly outperforming ResNet50 on validation metrics. Combining the two architectures through logit-weighted ensembling and test-time augmentation produced the strongest overall performance.

Despite these improvements, external evaluation on the ISIC 2019 dataset revealed a substantial performance drop compared to validation results, highlighting the impact of domain shift in medical image classification. Additionally, attempts to incorporate patient metadata reduced generalization performance, suggesting that structured features may introduce dataset-specific correlations that do not

transfer well across datasets. These findings underscore the importance of evaluating medical machine learning systems on independent datasets and carefully considering generalization when designing diagnostic models. Future work may explore domain adaptation techniques, larger multi-dataset training sets, or more robust multimodal architectures to improve cross-dataset performance.

7 Contributions

All group members contributed to each stage of the project. At the outset, we collectively agreed to use the ISIC 2020 public dataset and apply Convolutional Neural Networks for image classification. As a team, we conducted exploratory data analysis, examining both the images—which were the primary focus for our models—and the associated metadata. When developing and training our models, Artien and Kyle focused on implementing and tuning EfficientNet-B4, while Ann and Whitney worked on implementing baseline and ResNet50. This parallel approach allowed us to experiment with different preprocessing pipelines, as each architecture required different image scaling and padding strategies. As the project progressed, we iterated on our models and evaluation framework collaboratively. Artien took the lead in developing the final ensemble model, while the rest of the group provided feedback and assisted with testing and evaluation. Finally, all team members contributed to writing this report and organizing the GitHub repository for the final submission.

8 Reflections

At the beginning of the project our goal was to build two deep learning models—ResNet50 and EfficientNet-B4—and combine them into an ensemble in order to maximize predictive performance. We also aimed to explore whether incorporating patient metadata (such as demographic or anatomical site) alongside image data could further improve melanoma detection. However, during experimentation we found that including the metadata in our ensemble actually degraded model performance.

We have several hypotheses for why this may have occurred. One possible explanation relates to differences between datasets. Because the ISIC 2020 dataset does not provide publicly available ground-truth labels for its official test set, we instead evaluated our models on the ISIC 2019 dataset. It is possible that the patient populations or metadata distributions differ between these datasets, which may have introduced inconsistencies when incorporating structured features. Another possibility is that including metadata increased the complexity of the model and led to overfitting, particularly given the limited number of melanoma cases in the dataset.

Due to time constraints, we ultimately chose not to include patient metadata in the final ensemble and instead focused on optimizing the image-based models. Given additional time, we would further investigate multimodal modeling approaches that combine image features with structured patient information, as this more closely reflects how melanoma is evaluated in clinical practice. Additional experimentation with feature selection, regularization techniques, or alternative architectures for combining image and tabular data could potentially improve the effectiveness of this approach.

If we had more time, we would have investigated the compatibility between the ISIC 2019 and ISIC 2020 datasets earlier in the project. Because the ISIC 2020 dataset does not provide publicly available labels for its official test set, we relied on ISIC 2019 for cross-dataset evaluation. Earlier analysis of the differences between these datasets may have helped us better understand any performance issues during testing.

More broadly, we would be interested in further exploring how well different learning models trained on dermoscopic images generalize across datasets and under severe class imbalance conditions. One piece of feedback we received concerned the potential impact of patient demographics, particularly skin tone, on model performance. If the training data does not sufficiently represent diverse skin tones, models trained on this data may perform poorly on populations that were underrepresented during training. Unfortunately, race or skin tone was not included as part of the available metadata in the dataset, making it difficult to directly evaluate this issue. Investigating this question would likely require manual inspection or external annotation of a large number of images (30,000 images), which was beyond the scope of this project but represents an important area for future work.

9 Peer Review Responses

We appreciated the detailed feedback provided by our peers and found many of the suggestions helpful for reflecting on both the strengths and limitations of our approach. Several comments highlighted aspects of the project that we intentionally designed, while others identified areas that could be improved with additional time or experimentation.

One common theme in the feedback was the suggestion to incorporate patient metadata such as age, sex, and anatomical site in addition to the dermoscopic images. We agree that melanoma diagnosis is often informed by multiple factors, and multimodal modeling is a promising direction. While the dataset does contain some metadata fields, our initial goal was to focus on building strong image-based models using convolutional neural networks and evaluating how well these models could perform using

visual features alone. However, we did incorporate this feedback to see if a multimodal model would perform better, but it ended up performing worse.

Another point raised in the feedback relates to the severe class imbalance in the dataset, where melanoma cases represent roughly 2% of all samples. We attempted to address this issue through the use of a `WeightedRandomSampler` during training as well as specialized loss functions such as Focal Loss and `BCEWithLogitsLoss`. These methods were chosen to encourage the models to focus on the minority class rather than defaulting to predicting benign lesions. While these strategies did improve training stability, the reported precision-recall metrics suggest that detecting melanoma remains challenging. With additional time, we would explore more aggressive imbalance mitigation strategies such as stronger augmentation techniques, threshold tuning for classification decisions, or synthetic oversampling approaches.

The feedback also noted the performance drop observed during cross-domain evaluation, where models trained on the ISIC 2020 dataset were tested on images from the ISIC 2019 dataset. Although this decrease in performance is somewhat expected when evaluating models on data collected under different conditions, it does highlight the challenge of building models that generalize well across datasets. If we were to extend this work further, we would explore techniques aimed at improving generalization, such as domain-specific augmentation, training on combined datasets, or applying regularization methods that reduce overfitting to dataset-specific artifacts.

Finally, several reviewers commented on reporting consistency and experimental scope, including the absence of certain metrics and the mention of architectures such as Xception in the literature review without experimental results. These comments were helpful reminders of the importance of clear documentation. For the final submission, we ensured that model evaluation metrics were reported consistently across experiments and clarified which architectures were ultimately implemented within the project.

Overall, the feedback reinforced that our approach of combining a classical machine learning baseline with deep learning models was a valuable strategy, while also highlighting several opportunities for future work. In particular, integrating multimodal data, improving class imbalance handling, and strengthening cross-domain generalization represent promising directions that could further enhance the clinical relevance of this project.

References

- [1] A. R. Ali, J. Li, and S. J. O’Shea. Towards the automatic detection of skin lesion shape asymmetry, color variegation and diameter in dermoscopic images. *PLOS ONE*, 15(6):e0234352, 2020. doi: 10.1371/journal.pone.0234352.
- [2] Noel C. F. Codella, David Gutman, M. Emre Celebi, Brian Helba, Michael A. Marchetti, Stephen W. Dusza, Aadi Kalloo, Konstantinos Liopyris, Nabin Mishra, Harald Kittler, and Allan Halpern. Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (isbi), hosted by the international skin imaging collaboration (isic). *arXiv preprint arXiv:1710.05006*, 2017.
- [3] Andre Esteva, Brett Kuprel, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639):115–118, 2017. doi: 10.1038/nature21056.
- [4] Yassir Fathullah, Guoxuan Xia, and Mark Gales. Logit-based ensemble distribution distillation for robust autoregressive sequence uncertainties. *arXiv preprint arXiv:2305.XXXX*, 2023.
- [5] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [6] Carlos Hernández-Pérez, Marc Combalia, Srdjan Podlipnik, Noel C. Codella, Veronica Rotemberg, Allan C. Halpern, Omer Reiter, Cristina Carrera, Antonio Barreiro, Brian Helba, Susana Puig, Veronica Vilaplana, and Josep Malvehy. Bcn20000: Dermoscopic lesions in the wild. *Scientific Data*, 11(1):641, June 2024.
- [7] International Skin Imaging Collaboration. ISIC 2019 challenge dataset, 2019. Accessed via the ISIC Challenge Archive.
- [8] International Skin Imaging Collaboration. SIIM-ISIC 2020 challenge dataset, 2020. DOI: 10.34970/2020-ds01. Accessed via the ISIC archive.
- [9] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- [10] A. Madooei and M. S. Drew. Incorporating colour information for computer-aided diagnosis of melanoma from dermoscopy images: A retrospective survey and critical analysis. *International Journal of Biomedical Imaging*, 2016:4868305, 2016. doi: 10.1155/2016/4868305.
- [11] Matplotlib Development Team. Matplotlib. <https://matplotlib.org>, 2024.

- [12] Abid Mehmood, Yonis Gulzar, Qazi Mudassar Ilyas, Abdoh Jabbari, Muneer Ahmad, and Sajid Iqbal. Sbxception: A shallower and broader xception architecture for efficient classification of skin lesions. *Cancers*, 15(14):3604, 2023. doi: 10.3390/cancers15143604.
- [13] National Cancer Institute. Moles to melanoma: Recognizing the abcde features. <https://moles-melanoma-tool.cancer.gov/#/>, n.d. U.S. Department of Health and Human Services.
- [14] NumPy Developers. Numpy. <https://numpy.org>, 2024. Python library for numerical computing.
- [15] Pandas Development Team. pandas: Python data analysis library. <https://pandas.pydata.org>, 2024.
- [16] Pillow Contributors. Pillow: Python imaging library. <https://python-pillow.org>, 2024.
- [17] PyTorch Contributors. Pytorch. <https://pytorch.org>, 2024.
- [18] PyTorch Contributors. Torchvision. <https://pytorch.org/vision/stable/index.html>, 2024.
- [19] Veronica Rotemberg, Nicholas Kurtansky, Benjamin Betz-Stablein, Liam Caffery, Emmanouil Chousakos, Noel Codella, Marc Combalia, Stephen Dusza, Pascale Guitera, David Gutman, Allan Halpern, Brian Helba, Harald Kittler, Kivanc Kose, Stefan Langer, Kristina Lioprys, Josep Malvehy, Shaan Musthaq, Jagpreet Nanda, Omer Reiter, George Shih, Alexander Stratigos, Philipp Tschandl, Jonathon Weber, and H. Peter Soyer. A patient-centric dataset of images and metadata for identifying melanomas using clinical context. *Scientific Data*, 8:34, 2021. doi: 10.1038/s41597-021-00815-z.
- [20] Mayar Ashraf Saeed, Yasmine M. Afify, Nagwa Lotfy Badr, and Nivin A. Helal. Multimodal deep learning ensemble framework for skin cancer detection. *Scientific Reports*, 15:45660, 2025. doi: 10.1038/s41598-025-30534-z.
- [21] scikit-image Developers. scikit-image. <https://scikit-image.org>, 2024.
- [22] scikit-learn Developers. scikit-learn: Machine learning in python. <https://scikit-learn.org>, 2024.
- [23] SciPy Developers. Scipy. <https://scipy.org>, 2024.
- [24] Seaborn Contributors. Seaborn: Statistical data visualization. <https://seaborn.pydata.org>, 2024.
- [25] S. Seidenari, G. Pellacani, and C. Grana. Asymmetry in dermoscopic melanocytic lesion images: A computer description based on colour distribution. *Acta Dermato-Venereologica*, 86(2):123–128, 2006. doi: 10.2340/00015555-0043.
- [26] Kavitha Munuswamy Selvaraj, Sumathy Gnanagurusubbiah, Reena Roy Roby Roy, Jasmine Hephzipah John Peter, and Sarala Balu. Enhancing skin lesion classification with advanced deep learning ensemble models: A path towards accurate medical diagnostics. *Current Problems in Cancer*, 49:101077, 2024. doi: 10.1016/j.crrprobcancer.2024.101077.
- [27] Mingxing Tan and Quoc V. Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2019.
- [28] tqdm Developers. tqdm: A fast, extensible progress bar. <https://tqdm.github.io>, 2024.
- [29] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5:180161, 2018. doi: 10.1038/sdata.2018.161.
- [30] Ross Wightman. Pytorch image models (timm). <https://github.com/rwightman/pytorch-image-models>, 2024.